

STONY RAPIDS, SK, Canada

WMO: 711320

Lat: 59.250N

Lon: 105.840W

Elev: 801

StdP: 14.28

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.0	-37.1	-47.4	0.4	-41.1	-44.2	0.4	-36.8	16.9	-0.5	15.7	-3.8	1.3	100	0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	82.1	61.7	78.9	60.9	75.4	59.5	64.8	76.3	63.0	73.8	61.4	71.5	8.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	82.8	67.3	59.1	77.1	66.4	57.1	71.7	64.8	30.3	75.9	29.0	74.2	27.8	71.6	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.6	14.4	12.8		-45.8	88.1	4.4	3.2	-48.9	90.4	-51.5	92.3	-54.0	94.1	-57.1	96.4
			DB	-45.3	68.4	3.7	2.2	-47.9	70.0	-50.1	71.3	-52.1	72.6	-54.8	74.2
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.8	-11.1	-5.8	7.3	26.1	42.6	56.1	61.2	57.7	46.9	32.1	11.2	-4.7
	DBStd	28.01	15.62	14.24	15.55	13.39	9.39	7.05	5.45	6.07	7.56	8.65	13.38	15.66
	HDD50	9360	1895	1562	1325	717	264	25	1	9	146	555	1164	1697
	HDD65	13989	2360	1982	1790	1166	694	277	142	239	544	1019	1614	2162
	CDD50	891	0	0	0	1	36	206	347	246	53	2	0	0
	CDD65	45	0	0	0	0	1	9	24	11	0	0	0	0
	CDH74	1013	0	0	0	0	41	257	480	218	17	0	0	0
	CDH80	196	0	0	0	0	8	41	100	45	2	0	0	0
	WSAvg	5.3	4.7	4.9	5.6	6.1	6.4	5.8	5.2	4.9	5.4	5.2	4.8	4.3
	PrecAvg	15.7	0.6	0.6	0.7	0.6	1.0	1.5	2.4	2.9	2.2	1.3	0.9	0.5
(15) Precipitation	PrecMax	24.5	1.7	3.6	1.7	1.4	3.9	3.3	6.1	5.1	7.5	2.7	2.8	1.2
	PrecMin	10.2	0.0	0.0	0.0	0.0	0.1	0.0	1.2	0.7	0.7	0.4	0.1	0.0
	PrecStd	3.4	0.4	0.7	0.4	0.4	0.8	0.9	1.0	1.1	1.3	0.6	0.6	0.4
	PrecStd	3.4	0.4	0.7	0.4	0.4	0.8	0.9	1.0	1.1	1.3	0.6	0.6	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.9	31.9	47.2	63.1	79.0	84.4	86.8	85.7	76.2	60.5	39.7	30.4
		MCWB	29.5	30.3	37.6	47.3	56.8	61.5	64.7	65.3	60.6	50.5	35.7	28.9
	2%	DB	22.1	25.5	41.0	55.6	72.1	80.5	82.7	79.5	69.2	52.7	34.5	24.8
		MCWB	21.1	24.1	34.3	43.4	53.5	59.4	62.6	62.3	57.1	45.8	32.9	23.8
	5%	DB	17.2	20.6	35.2	50.6	66.6	76.9	79.5	75.6	64.1	47.8	32.1	21.1
		MCWB	16.4	19.4	31.0	40.5	50.9	58.5	61.7	61.1	53.9	43.0	30.6	20.2
	10%	DB	11.3	15.6	30.1	45.5	60.9	72.9	75.9	71.4	59.7	43.8	29.0	17.1
		MCWB	10.5	14.4	27.0	37.7	48.1	56.8	60.5	59.2	52.4	40.7	27.6	16.4
	0.4%	WB	29.7	30.4	38.2	48.1	58.4	65.6	67.6	67.2	61.8	51.6	36.2	29.2
		MCDB	30.1	31.2	46.5	61.3	74.7	75.9	80.0	80.3	74.1	57.9	38.5	30.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	21.3	24.1	34.6	43.9	54.9	62.5	65.3	64.3	58.1	46.3	33.1	23.8
		MCDB	22.2	25.3	40.1	54.3	68.7	74.4	77.0	76.0	66.0	51.3	34.4	24.7
	5%	WB	16.4	19.4	31.2	41.0	51.9	60.2	63.6	62.4	55.6	43.5	30.6	20.1
		MCDB	17.1	20.6	35.1	49.3	64.9	73.2	74.5	72.1	61.9	47.3	32.1	20.9
	10%	WB	10.6	14.5	27.0	38.1	49.0	58.1	62.0	60.4	53.3	40.8	27.6	16.3
		MCDB	11.3	15.6	30.1	44.9	59.7	70.3	72.3	68.7	58.9	43.7	28.9	16.9
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	20.5	24.9	23.5	23.0	23.7	23.2	20.9	17.9	12.2	13.7	15.4
		MCDBR	18.0	19.8	26.2	27.8	33.9	32.3	31.3	30.0	27.6	19.6	13.6	16.2
		MCWBR	17.7	18.9	20.1	17.2	17.7	14.7	14.1	14.6	15.7	13.1	12.3	15.9
		MCDBR	18.0	19.7	25.6	25.3	30.9	27.1	25.2	25.6	23.5	17.7	13.2	15.8
	5% WB	MCWBR	17.7	18.9	19.9	15.9	16.7	13.3	12.3	13.3	14.1	12.3	12.1	15.5
		MCWBR	17.7	18.9	19.9	15.9	16.7	13.3	12.3	13.3	14.1	12.3	12.1	15.5
(40) Clear-Sky Solar Irradiance	taub		0.214	0.254	0.259	0.322	0.330	0.364	0.403	0.358	0.310	0.286	0.230	0.182
	taud		2.166	2.230	2.303	2.226	2.436	2.396	2.317	2.434	2.588	2.520	2.216	2.083
	Ebn at Noon		200	247	284	279	286	276	260	266	266	238	188	171
	Edh at Noon		13	22	28	36	32	34	36	30	22	17	13	10
(44) All-Sky Solar Radiation	RadAvg		160	450	988	1559	1830	1885	1684	1308	836	386	135	88
	RadStd		14	35	60	109	184	166	95	84	91	41	17	10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page